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Revisited: An unsolved message from the Irish Republican Army (IRA)

Von [Klaus Schmeh](#) / 3. September 2018 / [14 Kommentare](#)

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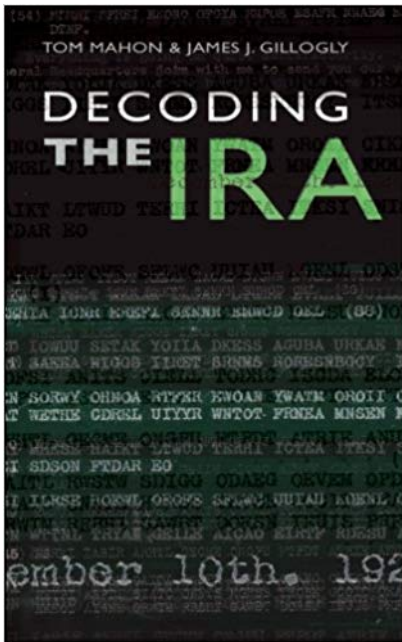
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US codebreaker Jim Gillogly broke over 1,000 transposition cryptograms created by IRA activists in the 1920s. Only one of these messages remained unsolved. Can a reader break it?



Decoding the IRA is the title of a book published in 2008 by codebreaking expert Jim Gillogly and historian Tom Mahon. This work is about 300 documents containing encrypted messages Mahon discovered in a Dublin archive. These documents originated from the estate of Irish activist Moss Twomey (1897-1978), who was the leader of the Irish Republican Army (IRA) from 1926 to 1936.



How Jim Gillogly decoded the IRA

Most of the encrypted texts are dispatches sent between the IRA headquarters in Dublin and IRA activists on the British Isles and in the USA. The rest of the collection consists of cipher messages exchanged between detained IRA members and Twomey or his comrades. In all, the corpus consists of about 1,300 individual cryptograms.

Tom Mahon, who researches the history of the IRA, lacked the expertise to decipher the cryptograms. He therefore asked the [American Cryptogram Association](#) (ACA) for help. When six of the cryptograms were published on the ACA mailing list, ACA member Jim Gillogly ...



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... immediately became interested. He analyzed the cryptograms and quickly solved all six of them. This was the start of a fruitful partnership between Jim and Tom Mahon. Over the next months, Jim managed to decipher almost all of the Twomey cryptograms. The cleartexts, which provided insights into the work of the IRA in the 1920s, proved extremely valuable for Tom Mahon's research project.

The first chapter of *Decoding the IRA*, in which Jim explains his deciphering work, is a fascinating read for everybody interested in codebreaking. The rest of the book, which is based on information gained from the deciphered messages, does not contain much information about cryptography, but it is worth reading for everybody interested in Irish history.

The following IRA message, which is dated 1927, is one of the six Tom Mahon sent to the ACA and that was subsequently shared on the ACA mailing list (it consists of 151 letters):

AEOOA IIIEO AEA EW LFRRD ELBAP RAE EA EIIIE AAAHO IFMFN COUMA
FSOSG NEGHS YPITT WUSYA ORDOO ERHNQ EEEVR TTRDI SOSDR ISIEE ISUTI
ERRAS TTKAH LFSUG RDLKP UEYDM ERNEO RULDC ERWTE ICNIA T

When analyzing this cryptogram, Jim Gillogly saw that the E was by far the most common letter with 23 appearances, followed by A, R and I. The letters Q, B and V turned out to be very rare. These frequencies are consistent with the English language, although the ratio of vowels (47 percent) seemed a little high (40 percent is usual). So, Jim assumed that he dealt with a transposition cipher. His guess was that a column-based transposition was used.

In his 40 years as a codebreaker, Jim Gillogly had written a considerable amount of cryptanalysis software for his personal use. Among other things, he was one of the first to use Hill Climbing for cryptanalysis. In this case, he used a Hill Climbing program tailored to break a column-based Transposition. He assumed a line length between 8 and 15 and started separate tries for each length. When he tried a line length of 12, he received the following cleartext candidate:

THEAADDARESTOWHECIEHYOUWILLOESENDSTUFFFOR...

This string contains many words that make sense. It is even possible to read a meaningful sentence (THE ADDRESS TO WHICH YOU WILL SEND STUFF ...) from it. However, there are quite a few letters that don't make sense. Jim restarted his program, with different starting keys, about a 100 times, but he didn't get a better result. The keyword his software found was FDBJALHCGKEI (this was certainly not the original one used by the IRA, but it was equivalent). For further analysis, he looked at the transposition table his program created:

FDBJALHCGKEI

THEAADDARESS
TOWHECIEHYOU
WILLOESENDST
UFFFOROQMGI
SMRSAWSEEENE
YFRUITDIERER
ANDGIERIENGR
OCERIIFIVEHA
ROLDECSEROSS
DUBLONIATRYT
OMAKAIEATUET
OAPPEAEARLLK

EFRHATI

Now, Jim immediately recognized how this result had come about: the encipherer had inserted two columns of meaningless vowels. In addition Jim's program had switched the L and the H column. So, here is the correct table:

FDBJALHCGKEI

THEA DD RESS
TOWH CI HYOU
WILL ES NDST
UFFF RO QMGI
SMRS WS EENE
YFRU TD ERER
ANDG ER ENGR
OCER IF VEHA
ROLD CS ROSS
DUBL NI TRYT
OMAK IE TUET
OAPP AE RLLK
EFRH TI

Here's the plaintext:

THE ADDRESS TO WHICH YOU WILL SEND STUFF FOR QMG IS MRS SWEENEY
FRUITERER AND GREENGROCER FIVE HAROLD'S CROSS DUBLIN TRY TO MAKE IT UP
TO APPEAR LIKE FRUIT.

An unsolved crypto mystery

Jim solved the other five cryptograms he received via the mailing list the same way. As it turned out, most of the 1,300 IRA cryptograms were encrypted in a column-based transposition. Jim could solve all of them – with only one exception:

GTHOO RCSNM EOTDE TAEDI NRAHE EBFNS INSGD AILLA YTTSE AOITDE.

The solution of this cryptogram is still unknown. Five years ago, I published [my first blog post about it](#) (in German), but noone could decipher it. A few readers remarked that the language of this message could be Gaelic instead of English. If you know more or if you can break the message, let me know.

Further reading: [The Top 50 unsolved encrypted messages: 24. The Erba murder cryptogram](#)

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[Kommentare \(14\)](#)

1. [#1](#) Thomas 
4. September 2018

Jim Gillogly provides sufficient information in his book (p. 33): “This message is identified as having fifty-two letters but only fifty-one appear in the cryptogram itself.” I wonder how he found out that the plaintext contains one additional letter which had been left out in the (most likely transposition) cryptogram.

2. [#2](#) Jerry McCarthy 
England, Europa
4. September 2018

Abo

3. [#3](#) George Lasry 
4. September 2018

I worked on this unsuccessfully, for example, by inserting a letter at every possible position (so did Jim as he wrote to me).

The number of letters is given in the message preamble.

4. [#4](#) Marc 
4. September 2018

There is a mistake in the solved cryptogram (151 letters), the word EXPECTED shouldn't be there 😊

5. [#5](#) Marc 
4. September 2018

I wonder if all transposition keys were 12 letters in length ?

6. [#6](#) Thomas 
4. September 2018

@Marc

So it is. Recovering the transposition keys, which were taken from Hawthorne's "The scarlett letter", is a fascinating story,

<https://books.google.de/books?id=gaNkC61RI18C&printsec=frontcover> (starting on p. 33).

7. [#7](#) Klaus Schmeh 
4. September 2018

Bart Wenmeckers via Facebook:

The crypto section of the book was interesting, well written and an enjoyable read.

8. [#8](#) Klaus Schmeh



4. September 2018

@Marc:

>I wonder if all transposition keys were

>12 letters in length ?

No, there are other lengths, as well.

9. [#9](#) Klaus Schmeh



4. September 2018

@Marc:

> the word EXPECTED shouldn't be there 😊

You're right, I corrected it.

10. [#10](#) Klaus Schmeh



10. September 2018

Richard Bean via Facebook:

When I got the book, I tried and failed, probably because my program was not very good, or the presence of dummy columns makes it very hard to distinguish between a good solution and the right solution. I did notice that Google Books has improved enough since 2008 to identify the source book of the keywords in Figure 8 as “Spiritualism: The Inside Truth” by Stuart C. Cumberland (1919). Jim probably knew that by now.

11. [#11](#) Klaus Schmeh



24. September 2018

Jim Gillogly via Facebook:

Yes, it was news to me. Good catch, Richard! The snippet search on the book allowed me to verify several of the phrases, so I'm confident you found the right book. Google Books is a bit of a tease in this case – I would think it would be in the public domain by now. From what I can tell the nearest copy of the book is about 1500 km away from me,, but perhaps I can find a library that will send copies of the relevant pages.



12. [#12](#) Richard Bean
Brisbane
6. August 2019

I've convinced Jim that the solution is ...

(choosing width 11, adding an L between the double E and forcing GELIGNIT, hexagram scoring using all the Project Gutenberg books I could scrape)

-2964330 REGELIGNITSCOTLANDSTAESTHEYRAIDEANDOBTAINEDOME OF THLS
CABJFDHGEKI 8 7 6 5 4 2 1 0 a: 2 0 1 9 5 3 7 6 4 10 8

Re Gelignit[e] Scotland sta[t]es they raide[d] and obtained [s]ome of th[i]s

I certainly like the "GELIGNIT" part – that's why I chose width 11 when I started seeing the highest scoring solutions have "LIGNIT", and that was scoring better than width 12. Very IRAish.

So it was hard to find the solution because of the errors – i.e. instead of dummy letters, this time, missing letters. Jim has a way of fixing these and using a width 12 matrix.



13. [#13](#) Jim Gillogly
6. August 2019

Yes, I believe this solution. Entering the columns of Richard's solution with an L added between the EE of the ciphertext, we see (if this were in a fixed-width font):

```
REGELIGNIT S
COTLANDSTA E
STHEYRAIDE A
NDOBTAINED O
MEOFTHLS
```

This indicates a width 12 cipher with the 11th column left out; 12 was a very common keyword/keyphrase length in this set of ciphers.

Checking the cipher transcription of the cipher on page 32 of my book, the L in THLS appears to be an overstrike of an I and L, so it's cromulent to conjecture that it's an I.

These are each credible errors: leaving out a letter (the L) while copying out the columns, and leaving out the whole 11th column, which should be ETDS between groups EOTDE and TAEDI if I've done it right. In essence these are the same error: sloppy copy from matrix to ciphertext.

Looks quite convincing to me.

For extra credit, figure out the 12-letter keyword or keyphrase – I haven't given it a serious try yet.



14. [#14](#) Richard Bean
Brisbane
8. August 2019

In the 37,912 Gutenberg books I have, the top 12-letter keyphrases which would work are AGAINILLSHOO (29 books), AGAINILLTHRO, (24 books) and DECISIONYETT (19 books).

I'll go for the first phrase and the first book which contains that phrase, book 1097, "Mrs Warren's Profession" by George Bernard Shaw (1902) because he's Irish and it's around the right time, and the most popular books in Project Gutenberg were digitised first (they're numbered sequentially).

<https://www.gutenberg.org/files/1097/1097-h/1097-h.htm>

FRANK [adroitly finishing the sentence for her] I should know why she is what she is, shouldn't I? What difference would that make?

Circumstances or no circumstances, Viv, you won't be able to stand your mother.

VIVIE [very angry] Why not?

FRANK. Because she's an old wretch, Viv. If you ever put your arm around her waist in my presence again, I'll shoot myself there and then as a protest against an exhibition which revolts me.

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Über den Autor



✕ @KlausSchmeh folgen

Klaus Schmeh zählt zu den weltweit führenden Experten für historische Verschlüsselungstechnik. Seine Bücher "Nicht zu knacken" (über die zehn größten ungelösten Verschlüsselungsrätsel) und "Codeknacker gegen Codemacher" (über die Geschichte der Verschlüsselungstechnik) sind Standardwerke. Auf "Cipherbrain" schreibt er über sein Lieblingsthema.

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